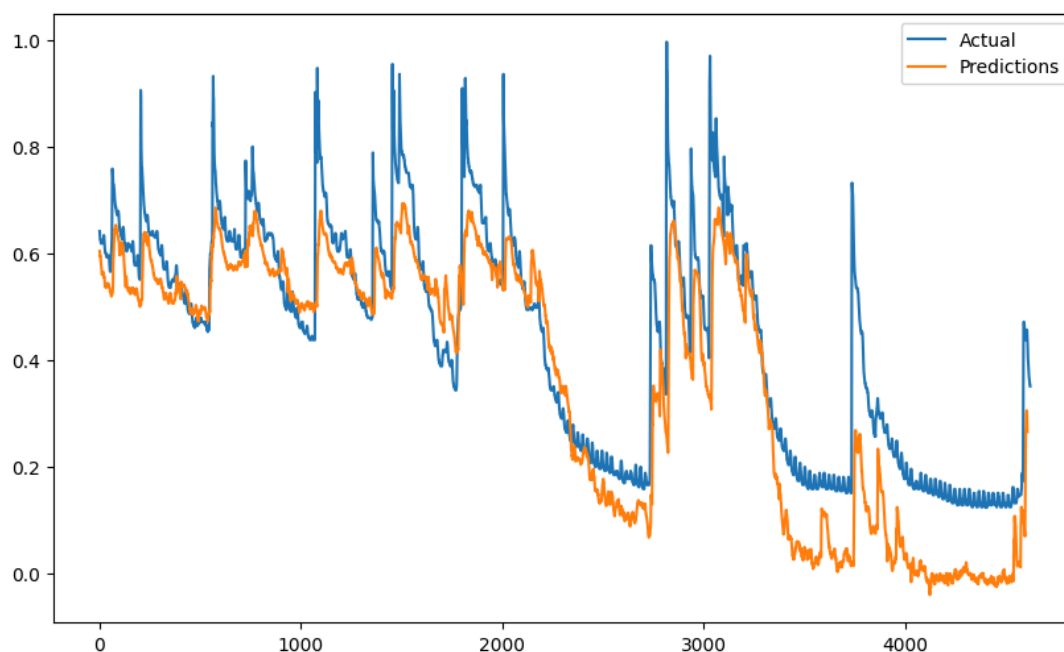


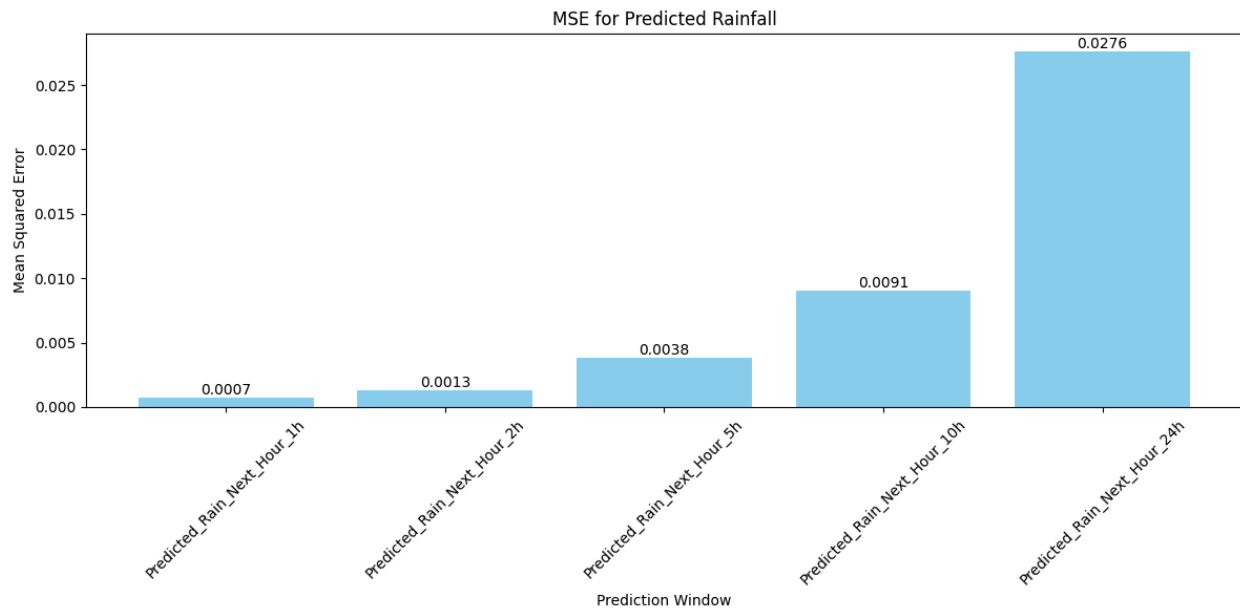
The weather prediction LSTM models were trained to predict SWC_5 given the feature engineered dataset for stations 1-6 using the feature engineering with rain chance calculator python notebook. The weather prediction folder has a folder for the feature engineered stations containing the datasets with the added columns with the past sums of 1 hour, 2 hours, 5 hours, 10 hours, and 24 hours, and then the predicted percentages for 1-7 days in the future for each hour of data. There were 2 ways created to approach creating the predicted percentages: creating a method to calculate the numbers based on different sums of past hours and then ranges over time and then scaling for future days vs creating a random first model to incorporate seasonality to analyze the data and then create new columns into the dataframe with the predicted future days.

	MSE	MAE	RMSE	MAPE
LSTM	0.027423	0.149868	0.165598	0.501657
BiLSTM	0.016400	0.081600	0.128063	0.191809
Feedback	0.018001	0.109814	0.134166	0.365493
BiFeedback	0.015512	0.084478	0.124548	0.253399

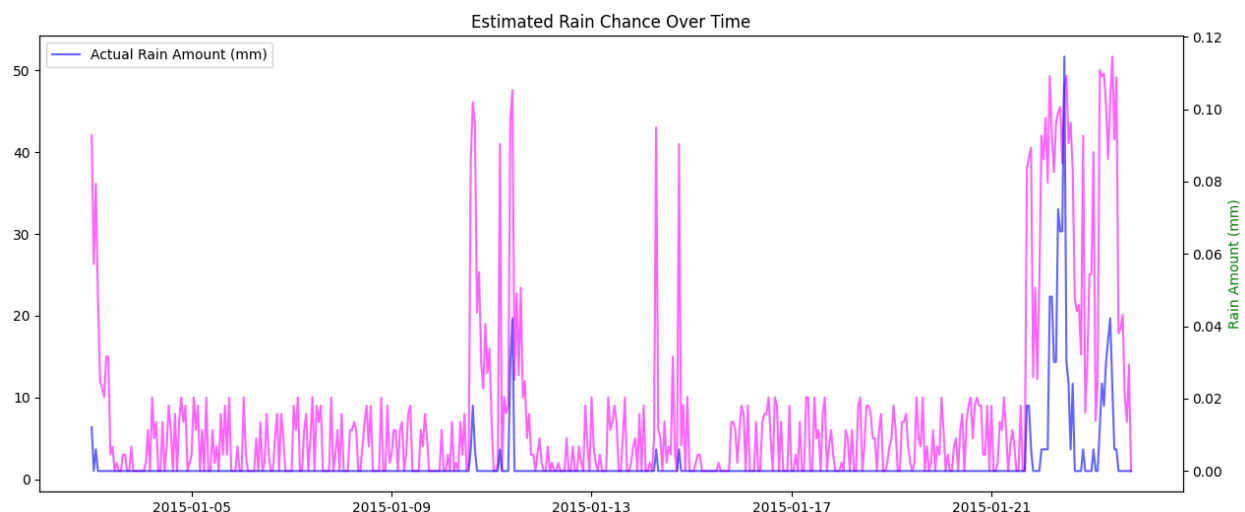
This is an example of the output of the first method using the bi_lstm_feedback_edited model made to run the newly created dataset on 4 different model LSTM, BiLSTM, Feedback, and BiFeedback. The metrics used to calculate error were MSE, MAE, RMSE, and MAPE.



This graph shows the actual versus predicted values from the chosen machine learning models over a range of the dates from the dataset. The blue line represents the actual values from the test dataset, while the orange line shows the predictions made by the 'Feedback' model. The x-axis represents the time steps of the data points, and the y-axis shows the values predicted by the model and the actual values observed.



This graph visualizes the mean squared error for predicted rainfall percentages for the next 1,2,5,10,and 24 hours using the past 1 hour to predict these values.



This graph visualizes the estimated rain chance over time on a subset of the data. The green and red values are the expected rainfall using the algorithm created by using past sums of data and the blue line is the actual rainfall. The Y axis on the left shows the

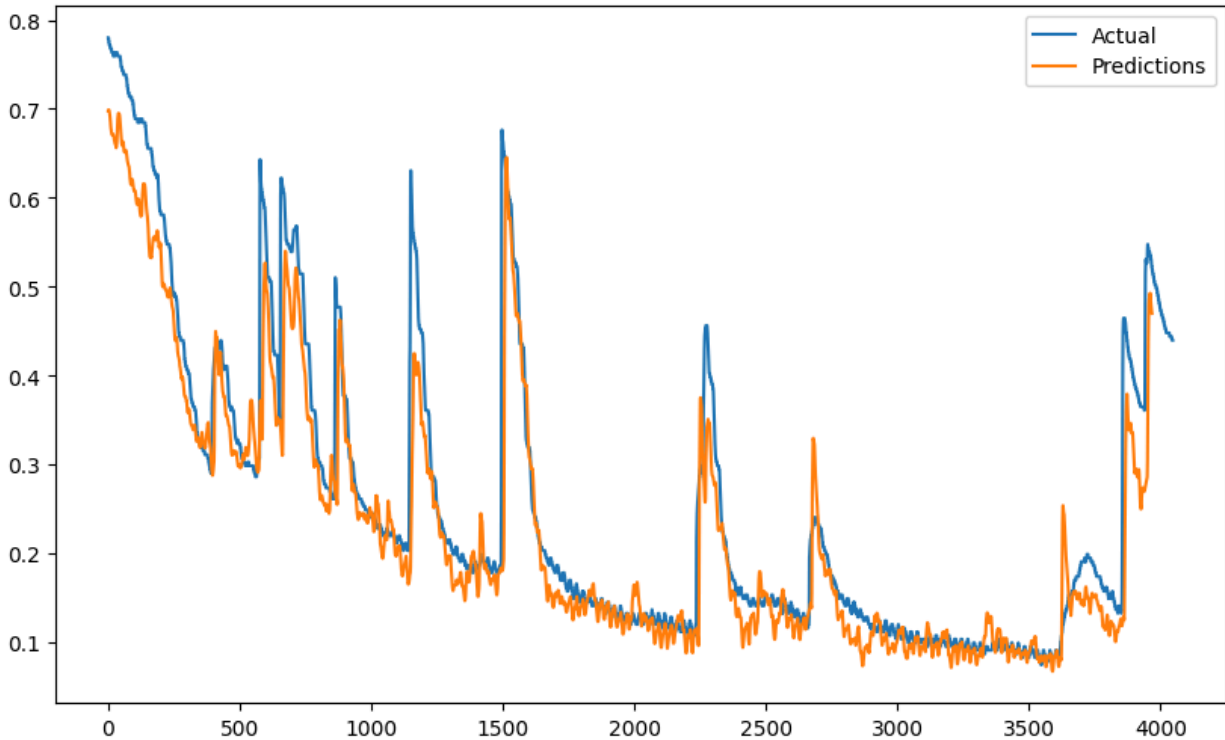
percentage of expected rainfall and the y axis on the right shows the amount of rainfall (mm).

Overall the method used for this approach was created by classifying rainfall into severity labels and assigning percentage values, which are dynamically adjusted based on recent rainfall data. These percentages are then methodically scaled to predict future rainfall probabilities, enhancing the accuracy of weather forecasts and soil moisture predictions. the algorithm based prediction over counts of different sum values for the past 24, 10 ,5 ,2 and 1 hours to take into account rainfall trends.

Now comparing this to the seasonality incorporated model, the error values were higher for the trained model according to these tables for station 2, meaning that the ranges algorithm worked slightly better for feature engineering the top table is the trained model feature engineering and the bottom is from the hardcoded ranges feature engineering.

	MSE	MAE	RMSE	MAPE
LSTM	0.036227	0.168650	0.190335	0.979781
BiLSTM	0.014202	0.073937	0.119170	0.250131
Feedback	0.018576	0.089387	0.136293	0.338286
BiFeedback	0.014165	0.078136	0.119016	0.283431

	MSE	MAE	RMSE	MAPE
LSTM	0.022402	0.126831	0.149674	0.365335
BiLSTM	0.015467	0.082170	0.124368	0.209130
Feedback	0.020088	0.119694	0.141731	0.426115
BiFeedback	0.024427	0.136963	0.156291	0.481686



This is an example of the predicted values from the chosen machine learning models over a range of the dates from the dataset for the trained model feature engineered dataset.

The approach for the trained model was to create a train/test batch of data from two different stations, run a random forest regressor and then use `y_pred` to predict the rainfall for the days in the future and let the regressor predict using `X_test`. The train features were 'SWC_5', 'SWC_10', 'SWC_20', 'SWC_50', 'T_5', 'T_10', 'T_20', 'T_50', 'Tair', 'Srad', 'Latitude', 'Longitude', 'Month', 'DayOfWeek', 'Predicted_Rain_Next_Hour_1h', 'Predicted_Rain_Next_Hour_2h', and they were used to train the model. Additionally, the `bi_lstm_comparison` model ran better with reducing the early stopping mechanism and increasing the number of epochs run.

Overall, both of the feature engineered datasets with the two different methods performed better at prediction precipitation and soil water content over the original dataset meaning that the added features of predicted precipitation values and past sums from shifting the data were key features in improving the accuracy of the model.